

EC400: Optimization and Fixed Points

Lecture 3: Comparative Statics and Fixed Points

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A Pre-Sessional Course

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Comparative Statics:

Envelope Theorem and Implicit Function Theorem

Envelope Theorem

Value Functions: Introduction

The value function of an optimization problem is the objective function evaluated at the solution.

Examples: profit functions, indirect utility functions, cost functions...

For a parameter b , consider a general constrained optimization problem:

$$\max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}|b) \text{ subject to } \mathbf{g}(\mathbf{x}|b) \leq \mathbf{0}.$$

For this constrained optimization problem:

- $\mathbf{g}(\mathbf{x}|b) = (g_1(\mathbf{x}|b), \dots, g_m(\mathbf{x}|b))^T$ denotes the constraints;
- $\mathbf{x}(b) = (x_1(b), \dots, x_n(b))^T$ denotes the optimal solution;
- $\boldsymbol{\mu}(b) = (\mu_1(b), \dots, \mu_m(b))$ denotes the corresponding multipliers;
- $L(\mathbf{x}, \boldsymbol{\mu}|b)$ denotes the Lagrangian of this problem;
- omit any slack constraints, so that $\boldsymbol{\mu}(b) \gg \mathbf{0}$.

Envelope Theorem

Definition

The **value function** of the problem amounts to

$$v(b) = f(\mathbf{x}(b)|b).$$

The Envelope Theorem relates changes in b to the value function.

Theorem

If $(\mathbf{x}(b), \mu(b))$ are differentiable and $\mathbf{x}(b)$ satisfies the relevant constraint qualification, then

$$\frac{dv(b)}{db} = \frac{\partial L(\mathbf{x}(b), \mu(b)|b)}{\partial b}.$$

Envelope Theorem: Proof

Proof of Theorem: Recall that FOC for the constraint optimization problem require that

$$\underbrace{D_{\mathbf{x}}f(\mathbf{x}(b)|b)}_{1 \times n} = \underbrace{\mu(b)}_{1 \times m} \underbrace{D_{\mathbf{x}}\mathbf{g}(\mathbf{x}(b)|b)}_{m \times n}.$$

Observe that, as all the constraints bind, total differentiation implies that

$$\underbrace{D_{\mathbf{x}}\mathbf{g}(\mathbf{x}(b)|b)}_{m \times n} \underbrace{D_b\mathbf{x}(b)}_{n \times 1} + \underbrace{D_b\mathbf{g}(\mathbf{x}(b)|b)}_{m \times 1} = \mathbf{0}.$$

Finally note that by partially differentiating the Lagrangian we get

$$\underbrace{D_bL(\mathbf{x}(b), \mu(b)|b)}_{1 \times 1} = \underbrace{D_bf(\mathbf{x}(b)|b)}_{1 \times 1} - \underbrace{\mu(b)}_{1 \times m} \underbrace{D_b\mathbf{g}(\mathbf{x}(b)|b)}_{m \times 1}.$$

... $\square$

Proof of Theorem: To prove the result observe that

$$\begin{aligned}\frac{dv(b)}{db} &= \frac{df(\mathbf{x}(b)|b)}{db} = D_b f(\mathbf{x}(b)|b) + D_{\mathbf{x}} f(\mathbf{x}(b)|b) D_b \mathbf{x}(b) = \\ &= D_b f(\mathbf{x}(b)|b) + \mu(b) D_{\mathbf{x}} \mathbf{g}(\mathbf{x}(b)|b) D_b \mathbf{x}(b) = \\ &= D_b f(\mathbf{x}(b)|b) - \mu(b) D_b \mathbf{g}(\mathbf{x}(b)|b) = \\ &= \frac{\partial L(\mathbf{x}(b), \mu(b)|b)}{\partial b}.\end{aligned}$$

where the first equality is definitional, the second holds by simple differentiation, the third holds by FOC, the fourth by totally differentiating constraints, and the fifth amounts to the partial of the Lagrangian. $\square$

Example: Interpreting the Lagrange Multiplier

Consider the problem:

$$\max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \leq b.$$

The Lagrangian of the problem is:

$$L(\mathbf{x}, \mu | b) = f(\mathbf{x}) - \mu(g(\mathbf{x}) - b).$$

The solution must satisfy for any b :

$$D_{\mathbf{x}}f(\mathbf{x}(b)) - \mu(b)D_{\mathbf{x}}g(\mathbf{x}(b)) = 0.$$

By total differentiation, since $g(\mathbf{x}(b)) = b$, we get:

$$D_{\mathbf{x}}g(\mathbf{x}(b))D_b\mathbf{x}(b) = 1.$$

Example: Interpreting the Lagrange Multiplier

By the chain rule and the previous observations, we get that:

$$\begin{aligned}\frac{df(\mathbf{x}(b))}{db} &= D_{\mathbf{x}}f(\mathbf{x}(b))D_b\mathbf{x}(b) \\ &= \mu(b)D_{\mathbf{x}}g(\mathbf{x}(b))D_b\mathbf{x}(b) \\ &= \mu(b).\end{aligned}$$

Thus, Lagrange multipliers tell us how much maximum value changes when the constraint is relaxed.

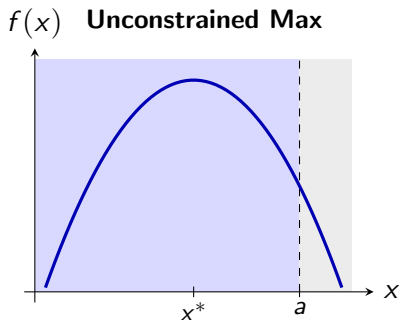
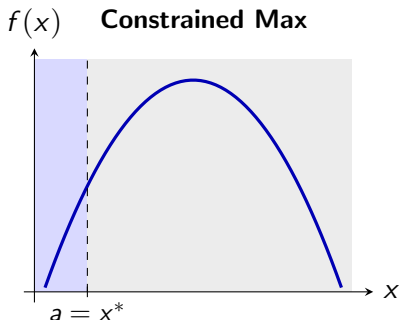
This is why they are called shadow prices, as they capture the value of violating a constraint.

Example: Interpreting the Lagrange Multiplier

The Lagrange multiplier can be interpreted as a **shadow price**.

For example, with profit maximization, Lagrange multipliers tell us how much maximum profits change when the constraint is relaxed.

They identify the most a firm would pay to acquire another unit of input.

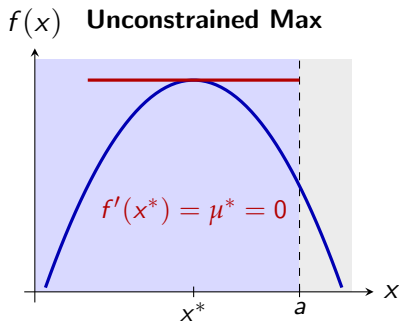
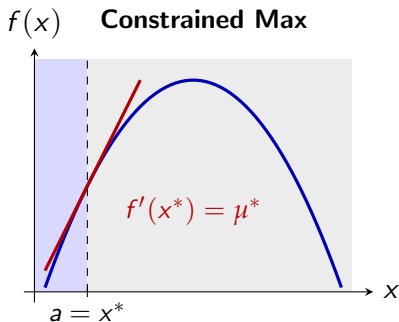


Example: Interpreting the Lagrange Multiplier

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Intuition: Unconstrained Problems

For a parameter b , consider the unconstrained optimization problem:

$$\max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}|b).$$

Let $\mathbf{x}(b)$ be the solution, and let it be continuous and differentiable in b .

Since $\partial f(\mathbf{x}(b)|b)/\partial \mathbf{x} = \mathbf{0}$ by FOC, the envelope theorem simplifies to

$$\frac{df(\mathbf{x}(b)|b)}{db} = \frac{\partial f(\mathbf{x}(b)|b)}{\partial \mathbf{x}} \frac{\partial \mathbf{x}(b)}{\partial b} + \frac{\partial f(\mathbf{x}(b)|b)}{\partial b} = \frac{\partial f(\mathbf{x}(b)|b)}{\partial b}.$$

Intuition: At a maximum, changing slightly the parameters does not affect the value through changes in the solution $\mathbf{x}(b)$, because FOC hold. Thus, only the partial derivative matters. If there are multiple solutions, you have to make sure that you do not jump to a different solution.

Implicit Function Theorem

Implicit Function Theorem: Intro

Many relevant economic problems study how an equilibrium or a solution to an optimization problem is affected by exogenous variables.

The **Implicit Function Theorem** IFT is the main tool for such comparisons.

The IFT provides sufficient conditions for a set of simultaneous equations

$$F_i(\mathbf{x}|\mathbf{b}) = F_i(x_1, \dots, x_n | b_1, \dots, b_m) = 0 \text{ for } \forall i = 1, \dots, n \text{ or } \mathbf{F}(\mathbf{x}|\mathbf{b}) = \mathbf{0},$$

to have a solution defined by **implicit functions**

$$x_i = x_i(\mathbf{b}) = x_i(b_1, \dots, b_m) \text{ for } \forall i = 1, \dots, n \text{ or } \mathbf{x} = \mathbf{x}(\mathbf{b}).$$

IFT ensures that the n equations defining the system can be solved in the n variables, $x_1, \dots, x_n$ even when explicit form solutions do not exist.

Implicit Function Theorem: Determinant of the System

Consider the determinant of the $n \times n$ Jacobian of the system $\mathbf{F}$ with respect to the $\mathbf{x}$ variables:

$$|D_{\mathbf{x}}\mathbf{F}(\mathbf{x}|\mathbf{b})| = \left| \frac{\partial \mathbf{F}(\mathbf{x}|\mathbf{b})}{\partial \mathbf{x}} \right| = \begin{vmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial F_n}{\partial x_1} & \frac{\partial F_n}{\partial x_2} & \cdots & \frac{\partial F_n}{\partial x_n} \end{vmatrix}.$$

Implicit functions, $\mathbf{x}(\mathbf{b})$, exist only if this determinant differs from 0.

Intuitively, such functions exist when the system can be approximated by a linear system of equations.

Implicit Function Theorem

Theorem (Implicit Function Theorem)

Consider the system of equations $\mathbf{F}(\mathbf{x}|\mathbf{b}) = \mathbf{0}$. Let any function F_i have continuous partial derivatives with respect to all $\mathbf{b}$ and $\mathbf{x}$ variables.

Consider a point $(\bar{\mathbf{x}}, \bar{\mathbf{b}})$ such that $\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}}) = \mathbf{0}$ and:

$$|D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})| \neq 0.$$

If so, then the following hold:

- *there exists an m -dimensional neighborhood of $\bar{\mathbf{b}}$ in which the variables $\mathbf{x}$ are functions of $\mathbf{b}$;*
- *for every vector $\mathbf{b}$ in the neighborhood $\mathbf{F}(\mathbf{x}(\mathbf{b})|\mathbf{b}) = \mathbf{0}$;*
- *the implicit functions $\mathbf{x}(\mathbf{b})$ are continuous and have continuous partial derivatives with respect to all the variables $\mathbf{b}$.*

Derivatives of Implicit Functions

IFT allows us to find the partial derivatives of the implicit functions $\mathbf{x}(\mathbf{b})$ without solving for the $\mathbf{x}$ variables.

To do so, since in the neighborhood of the solution the equations have a status of identities, we can set the total differential to zero,

$$\frac{dF(\mathbf{x}(\mathbf{b})|\mathbf{b})}{d\mathbf{b}} = 0.$$

For instance, when considering only $db_i \neq 0$ and setting $db_j = 0$ for $j \neq i$, the result in matrix notation implies that

$$\frac{\partial \mathbf{F}(\mathbf{x}(\mathbf{b})|\mathbf{b})}{\partial \mathbf{x}} \frac{\partial \mathbf{x}(\mathbf{b})}{\partial b_i} + \frac{\partial \mathbf{F}(\mathbf{x}(\mathbf{b})|\mathbf{b})}{\partial b_i} = 0 \quad \text{implies}$$

$$\frac{\partial \mathbf{x}(\mathbf{b})}{\partial b_i} = - \left[\frac{\partial \mathbf{F}(\mathbf{x}(\mathbf{b})|\mathbf{b})}{\partial \mathbf{x}} \right]^{-1} \frac{\partial \mathbf{F}(\mathbf{x}(\mathbf{b})|\mathbf{b})}{\partial b_i}.$$

Derivatives of Implicit Functions

Since $|D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})| \neq \mathbf{0}$ there a unique solution exists to this linear system.

By Cramer's rule, such a solution equals

$$\frac{\partial x_j(\mathbf{b})}{\partial b_i} = \frac{|D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})_j|}{|D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})|}$$

where $D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})_j$ is the matrix formed by replacing the j^{th} column of $D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})$ with the vector $-D_{b_i}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})$.

Implicit Function Theorem: Example

For $n = 2$ and $m = 1$, consider the system of equations

$$\mathbf{F}(\mathbf{x}|b) = \begin{cases} x_1 - e^{x_2} \\ x_1^2 + x_2^2 - b \end{cases} = \mathbf{0}.$$

Restrict attention to the case in which $\mathbf{x} \geq \mathbf{0}$.

When $\mathbf{x} \neq \mathbf{0}$, we have that

$$|D_{\mathbf{x}}\mathbf{F}(\mathbf{x}|b)| = 2x_2 + 2x_1e^{x_2} > 0.$$

If so, IFT applies and there exists implicit functions such that

$$\mathbf{F}(\mathbf{x}(b)|b) = \begin{cases} x_1(b) - e^{x_2(b)} \\ x_1(b)^2 + x_2(b)^2 - b \end{cases} = \mathbf{0}$$

Implicit Function Theorem: Example

By totally differentiating the system, we find

$$\frac{d\mathbf{F}(\mathbf{x}(b)|b)}{db} = \begin{cases} x_1'(b) - x_2'(b)e^{x_2(b)} \\ 2x_1'(b)x_1(b) - 2x_2'(b)x_2(b) - 1 \end{cases} = \mathbf{0}.$$

By solving the linear system and requiring $\mathbf{x} = \mathbf{f}(b)$, we find that

$$\frac{\partial \mathbf{x}(b)}{\partial b} = - \left[\frac{\partial \mathbf{F}(\mathbf{x}(b)|b)}{\partial \mathbf{x}} \right]^{-1} \frac{\partial \mathbf{F}(\mathbf{x}(b)|b)}{\partial b} = \begin{cases} e^{x_2} / (2x_2 + 2x_1 e^{x_2}) \\ 1 / (2x_2 + 2x_1 e^{x_2}) \end{cases}.$$

We **do not solve the non-linear system $\mathbf{F}$ to find the slopes** of the implicit functions, we only solve a linear system of partials!

Implicit Function Theorem: Comparative Statics

If we apply the IFT in the context of an optimization problem:

- $\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}}) = \mathbf{0}$ amounts to a system of FOC and binding constraints.
- $D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})$ amounts to the bordered Hessian (or the Hessian of the Lagrangian obtained by deleting slack constraints).
- $|D_{\mathbf{x}}\mathbf{F}(\bar{\mathbf{x}}|\bar{\mathbf{b}})| \neq \mathbf{0}$ in any strict local optimum.

In other words, we can take a set of equilibrium conditions, totally differentiate them and find out how endogenous variables respond to changes in exogenous ones in a neighbourhood of the solution.

Implicit Function Theorem: Comparative Statics

Consider an optimization with two variables and one equality constraint:

$$\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) \leq b.$$

First order conditions require

$$F^1(\mu, \mathbf{x}|b) = -g(\mathbf{x}) + b = 0,$$

$$F^2(\mu, \mathbf{x}|b) = f_1(\mathbf{x}) - \mu g_1(\mathbf{x}) = 0,$$

$$F^3(\mu, \mathbf{x}|b) = f_2(\mathbf{x}) - \mu g_2(\mathbf{x}) = 0.$$

We need to verify that the Jacobian of the system is different from zero,

$$|D_{\mu, \mathbf{x}} \mathbf{F}(\mu, \mathbf{x}|b)| = \begin{vmatrix} 0 & -g_1 & -g_2 \\ -g_1 & f_{11} - \mu g_{11} & f_{21} - \mu g_{21} \\ -g_2 & f_{12} - \mu g_{12} & f_{22} - \mu g_{22} \end{vmatrix} \neq 0.$$

But this matrix simply amounts to the bordered Hessian of the problem.

Implicit Function Theorem: Comparative Statics

Thus, when strict second order conditions are satisfied, we have that:

- 1 The determinant of the bordered Hessian must differ from zero, (it is negative definite).
- 2 By totally differentiating FOC, we find how a solution changes with b

$$\begin{aligned}g_1 \frac{\partial x_1}{\partial b} + g_2 \frac{\partial x_2}{\partial b} - 1 &= 0, \\(f_{11} - \mu g_{11}) \frac{\partial x_1}{\partial b} + (f_{12} - \mu g_{12}) \frac{\partial x_2}{\partial b} - g_1 \frac{\partial \mu}{\partial b} &= 0, \\(f_{21} - \mu g_{21}) \frac{\partial x_1}{\partial b} + (f_{22} - \mu g_{22}) \frac{\partial x_2}{\partial b} - g_2 \frac{\partial \mu}{\partial b} &= 0.\end{aligned}$$

At the solution, one can then solve for:

$$\frac{\partial x_1}{\partial b}, \frac{\partial x_2}{\partial b} \text{ and } \frac{\partial \mu}{\partial b}.$$

Correspondences:

Fixed Point Theorem

Correspondences

As equilibria are generally defined as fixed points, proving equilibrium existence essentially relies on fixed point theorems.

We state results more generally for correspondences as the solutions of the optimization problems may not be unique.

Definition (Correspondence)

A **correspondence** from set X to set Y , denoted $h : X \rightrightarrows Y$, is a function from X to $P(Y) = \{A | A \subseteq Y\}$ which is the set of all subsets of Y .

A correspondence is **non-empty** if $h(x) \neq \emptyset$ for all $x \in X$.

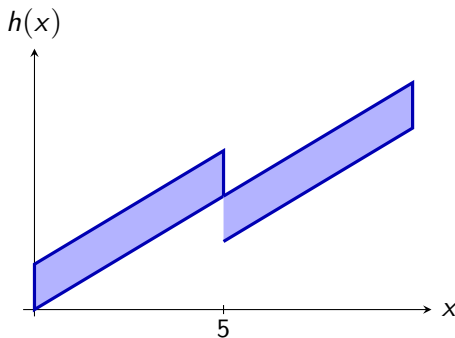
Remark

A correspondence h is set-valued function, as $h(x) \subseteq Y$ for every $x \in X$.

Examples: Correspondences

For instance the following set-valued functions are correspondences:

- $h(x) = [x, x + 2]$ for all $x \in [0, 5]$;
- $h(x) = [x - 2, x]$ for all $x \in (5, 10]$.



Note: Closed boundaries are drawn as solid lines.

Properties of Correspondences

Definition (Properties of Correspondences)

A correspondence $h : X \rightrightarrows Y$ is:

convex-valued if $y \in h(x)$ and $y' \in h(x)$ imply that

$$py + (1 - p)y' \in h(x) \text{ for any } p \in [0, 1];$$

upper-hemicontinuous when Y is compact if for any a sequence $\{y^k, x^k\}_{k=1}^{\infty}$ such that $y^k \in h(x^k)$ for all k ,

$$x^k \rightarrow x \text{ and } y^k \rightarrow y \text{ imply that } y \in h(x).$$

Remark

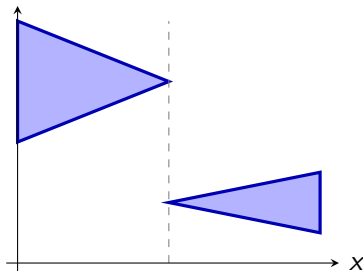
Any function is a convex-valued correspondence.

Any continuous function is a upper-hemicontinuous correspondence.

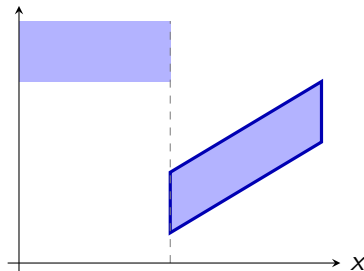
Properties of Correspondences

The following plots highlight the properties defined earlier:

$h(x)$ **UHC but not Convex**



$h(x)$ **Convex but not UHC**



Remark

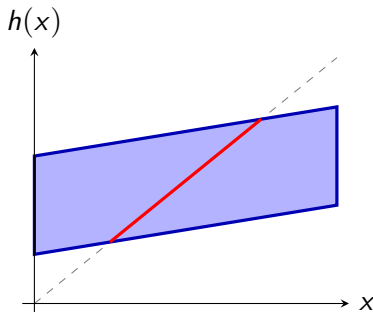
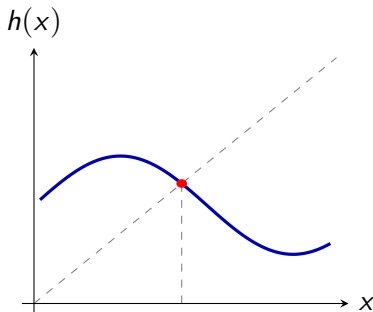
A correspondence h with compact codomain Y is upper-hemicontinuous if and only if its graph $Gr(h) = \{(y, x) \in Y \times X \mid y \in h(x)\}$ is closed.

Fixed Points of a Correspondence

Definition (Fixed Point)

A **fixed point** of the correspondence is a point $x \in X$ such that $x \in h(x)$.

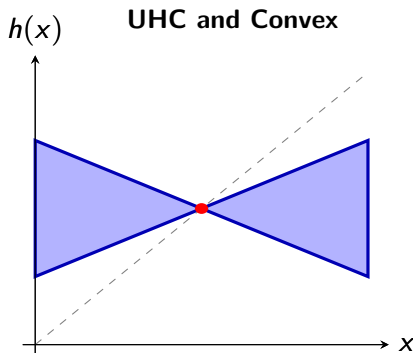
The following plots display fixed points in red:



Fixed Point Theorem

Theorem (Kakutani Fixed Point Theorem)

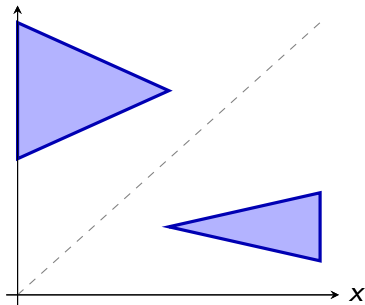
Let X be a non-empty, compact and convex subset of $\mathbb{R}^n$. Let $h : X \rightrightarrows X$ be a non-empty, convex-valued, upper-hemicontinuous correspondence. Then h has a fixed point.



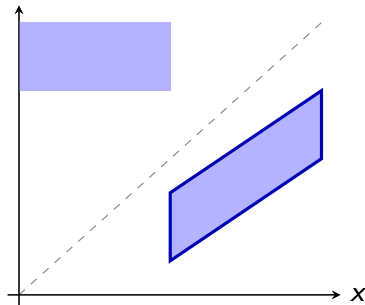
Fixed Point Theorem: Why UHC and Convex?

The plots show why the correspondence had to be upper-hemicontinuous and convex-valued for a fixed point theorem to apply:

$h(x)$ **UHC but not Convex**



$h(x)$ **Convex but not UHC**



Intuition: Fixed Point Theorem

Let $X = [0, 1]$ which is a non-empty, compact and convex.

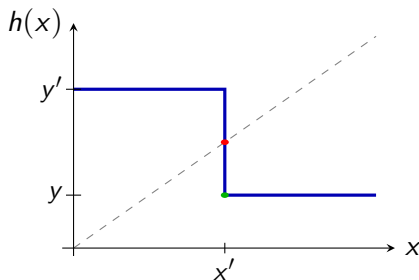
If $0 \in h(0)$ or if $1 \in h(1)$, we have a fixed point.

If not, $y > 0$ for all $y \in h(0)$ and $y < 1$ for all $y \in h(1)$.

If so, take the smallest value x' such that $y \leq x'$ for some $y \in h(x')$.

By UHC, there exists $y' \in h(x')$ such that $y' \geq x'$.

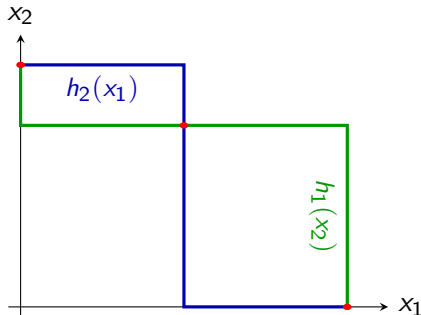
But as h is convex-valued, we have that $x' \in h(x')$.



Intuition: Fixed Points Two Dimensions

Consider a two-dimensional correspondence $h(x) \subseteq \mathbb{R}^2$.

If $h_i(x) = h_i(x_j)$, we can plot the fixed points in the following graph.



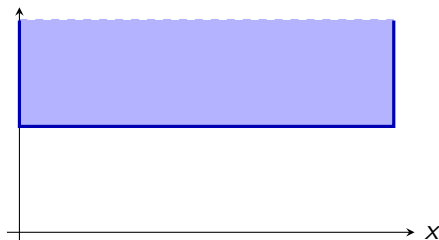
More Properties of Correspondences

Definition (Properties of Correspondences)

A correspondence $h : X \rightrightarrows Y$ is:

compact-valued if $h(x)$ is compact for all $x \in X$.

$h(x)$ **UHC but not Compact**



Example: $h(x) = [k, \infty)$ for all $x \in [0, 1]$.

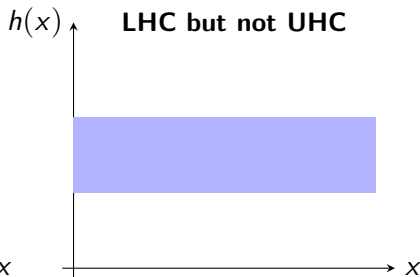
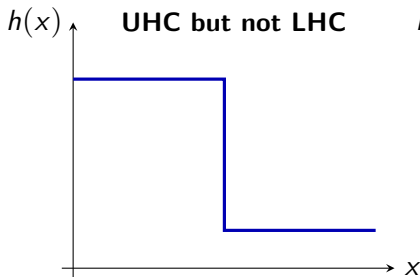
More Properties of Correspondences

Definition (Properties of Correspondences)

A correspondence $h : X \rightrightarrows Y$ is:

lower-hemicontinuous if for any $(y, x) \in \text{Gr}(h)$ and for any sequence $\{x^k\}_{k=1}^{\infty}$ such that $x^k \rightarrow x$, there exists a sequence $\{y^k\}_{k=1}^{\infty}$ such that

$$y^k \in h(x^k) \text{ for all } k \text{ and } y^k \rightarrow y.$$



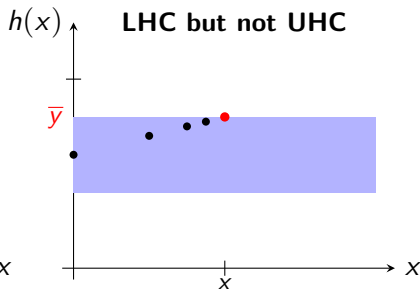
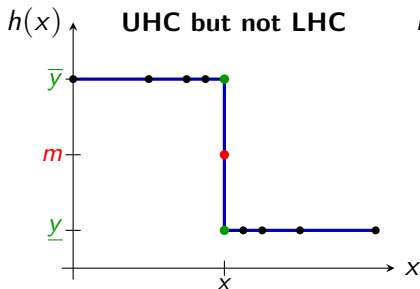
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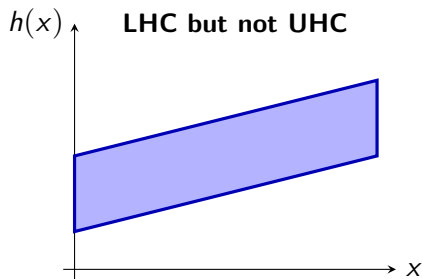


More Properties of Correspondences

Definition (Properties of Correspondences)

A correspondence $h : X \rightrightarrows Y$ is:

continuous if it is both lower-hemicontinuous and upper-hemicontinuous.



Example: $h(x) = [(1 + x)/4, (2 + x)/4]$ for all $x \in [0, 1]$.

Correspondences:

Theorem of Maximum

Theorem of the Maximum

Theorem (Theorem of the Maximum)

Let X be a compact subset of $\mathbb{R}^n$, and let B be a subset of $\mathbb{R}^m$.

Let $f : X \times B \rightarrow \mathbb{R}$ be a continuous function.

Let $g : B \rightrightarrows X$ be a continuous correspondence.

Let $x^* : B \rightrightarrows X$ be the correspondence defined by

$$x^*(\mathbf{b}) = \arg \max_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}|\mathbf{b}) \text{ subject to } \mathbf{x} \in g(\mathbf{b}).$$

If so, the following must hold:

- (1) x^* is *upper-hemicontinuous*;
- (2) if g is *non-empty and compact-valued*, so is x^* .

The theorem plays a key role in invoking FPT as it delivers UHC and non-empty compact values (but *not convex values*).

Theorem of the Maximum

If one further invokes that:

- $f(\mathbf{x}|\mathbf{b})$ is quasi-concave in $\mathbf{x}$ for all $\mathbf{b}$;
- $g(\mathbf{b})$ convex-valued for all $\mathbf{b}$;

then x^* is **convex-valued**.

These conditions will be critical to prove equilibrium existence.

Theorem Of the Maximum: Applications

Best Response: Let player i have utility $u_i(s_i, s_{-i})$ and strategy set S_i . Define:

$$\beta_i(s_{-i}) = \arg \max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

- If u_i is **continuous** and S_i compact, β_i is UHC and compact-valued.
- If u_i is **linear** and S_i is convex, β_i is convex-valued (mix of BRs is BR).
- If u_i is **strictly concave** in s_i , β_i is single-valued.

Marshallian Demand: Let $u(x)$ be utility, $B(p, m) = \{x \geq 0 \mid px \leq m\}$ be the compact and convex budget set. Define:

$$x^*(p, m) = \arg \max_{x \in B(p, m)} u(x).$$

- If u is continuous, x^* is UHC and compact-valued.
- If u is **quasi-concave**, x^* is convex-valued.
- If u is **strictly quasi-concave**: x^* is single-valued.

Extra: A Simple Dynamic Problem

Extra: A Dynamic Consumption-Saving Problem

A player starts life with wealth $a_0 > 0$.

Each period $t = 0, \dots, \infty$, he decides how much to consume $c_t \geq 0$.

Consumption is chosen to maximize your lifetime utility,

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in [0, 1)$ is the discount factor.

Assume that $\lim_{c \rightarrow 0} u'(c) = +\infty$, so that $c_t > 0$ each period.

Wealth at the beginning of period t is denoted by a_t .

Wealth a_t is invested at a constant interest rate r .

Wealth evolves according to the law of motion

$$a_{t+1} = (1 + r)a_t - c_t.$$

Extra: A Dynamic Consumption-Saving Problem

Given $a_0 > 0$, the problem is solved by $\{c_t\}_{t=0}^{\infty}$ and $\{a_t\}_{t=0}^{\infty}$ to maximize

$$\sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1)$$

subject to the constraints that for all $t \geq 0$,

$$a_{t+1} = (1 + r)a_t - c_t, \quad (2)$$

$$a_{t+1} \geq 0. \quad (3)$$

To find the sequences $(\mathbf{c}, \mathbf{a}) = \{c_t, a_{t+1}\}_{t=0}^{\infty}$ write the Lagrangian:

$$L(\mathbf{c}, \mathbf{a} | \boldsymbol{\mu}, \boldsymbol{\lambda}) = \sum_{t=0}^{\infty} [\beta^t u(c_t) - \mu_t(a_{t+1} - (1 + r)a_t + c_t) + \lambda_t a_{t+1}].$$

where $(\boldsymbol{\mu}, \boldsymbol{\lambda}) = \{\mu_t, \lambda_t\}_{t=0}^{\infty}$ are sequences of Lagrange multipliers associated to the sequences of constraints (2) and (3).

Extra: A Dynamic Consumption-Saving Problem

The first-order conditions of the problem require for all $t \geq 0$:

$$\begin{aligned}\partial L / \partial c_t &= \beta^t u'(c_t) - \mu_t = 0, \\ \partial L / \partial a_{t+1} &= (1+r)\mu_{t+1} - \mu_t + \lambda_t = 0.\end{aligned}$$

Complementary slackness requires instead for all $t \geq 0$:

$$\lambda_t a_{t+1} = 0.$$

Necessary conditions are also sufficient if the problem is well behaved.

Wealth a_t never falls to zero. If $a_T = 0$ for some T , then $c_t = 0$ for all $t \geq T$, which cannot be as $u'(0) = \infty$.

Extra: A Dynamic Consumption-Saving Problem

Therefore, we have that for all $t \geq 0$:

- $a_{t+1} > 0$;
- $\lambda_t = 0$;
- $(1 + r)\mu_{t+1} = \mu_t$.

Consequently, for all $t \geq 0$, consumption c_t satisfies:

$$u'(c_t) = (1 + r)\beta u'(c_{t+1}),$$

and, if so, wealth a_t satisfies for all $t \geq 0$:

$$a_{t+1} = (1 + r)a_t - c_t.$$